

# AI-Driven Multilingual Dubbing Framework for Awareness and Educational Videos

Mouna M<sup>1\*</sup>  
Presidency University,  
Bangalore

MOUNA.20231CSE0375@presidencyuniversity.in

Priyanka Savadekar<sup>2</sup>  
Presidency University,  
Bangalore

priyanka@presidencyuniversity.in

Debasmita Mishra<sup>3</sup>  
Presidency University,  
Bangalore

debasmita.mishra@presidencyuniversity.in

Neha Arora<sup>4</sup>  
Presidency University,  
Bangalore

neha.arora@presidencyuniversity.in

Srabana Pramanik<sup>5</sup>  
Presidency University,  
Bangalore

srabana.pramanik@presidencyuniversity.in

Amreen Khanum D<sup>6</sup>  
Presidency University,  
Bangalore

amreen.khanum@presidencyuniversity.in

**Abstract**—Translation is the process of being a bridge between two people who speak different languages. One person says their thoughts, emotions, and express it in another language so the other person can truly understand it. This paper describes the system which translate awareness videos from English to other regional languages. India being a diverse country with many religions, diverse culture, tradition and languages. It is important to provide awareness to people and reach out every part of our country and reach our thoughts to the people around the country. The regional languages used in this work are Hindi, Kannada, Telugu, Malayalam and Tamil. The translation process here uses raspberry pi, SD CARD and a speaker. Here our work uses raspberry pi as our key component. The primary objective of the system is to enhance the accessibility and effectiveness of educational and awareness videos by overcoming language barriers. This system is particularly beneficial for diverse linguistic regions such as Africa, Uganda, South Korea, and India, where language diversity can often hinder the dissemination of important messages. English being a universal language is available worldwide but many people across the world face difficulty in understanding English or learning English and this software will help people to understand their videos in regional languages.

**Keywords**— *Speech Recognition, Machine Translation, Speech Synthesis, Artificial Intelligence*

## I. INTRODUCTION

India is a diverse country with 28 states with more than 270 languages. Developing a software to Translate English awareness videos into regional languages such as Kannada, Telugu, and Tamil Etc. This article comments are used to promote communication between people, it should give confidence and accuracy section. The speech translation process is to convert the speech phrase from one language to another language. The speech translation is used to make the human communication Better [1]. This software could help people to bring awareness among themselves as well as the society. Here raspberry pi will survive as a key component for our system. Voice assistants like Siri, Google Assistant, and Alexa are widely used examples of automation today, but their functionality is currently quite limited. To overcome these limitations, a more advanced voice command system could be developed to act as an intelligent personal assistant. This system would be capable of performing a broader range

of tasks and services for users, thereby validating and enhancing the usefulness of voice assistants. Furthermore, the limited accessibility of existing voice assistant technologies motivates the development of the proposed prototype to enable wider adoption across diverse regions both nationally and globally. In a world where effective communication is crucial, language barriers often hinder the dissemination of important information. This software will leverage speech recognition and translation of videos. It makes sure that it provides efficient and accurate translation of videos. Making the videos accessible worldwide in diverse languages. Our system enhances efficiency, flexibility and ease of use to the people specially who live in these parts of the world Africa, Uganda, South Korea, and India. It also helps to educate people across the world with basic awareness. And teaches people to get adapt to the new technologies making technology an easier adaptable aspect to life. The motivation to our system goes down as below.

The main aim of our system is to develop a software for dubbing awareness videos into regional languages. This system will help reduce the communication barriers and help with an information source to be disseminated throughout the community. In many areas, language can hinder individuals from attaining information or resources related to educational or other knowledge which may impact their ability to make informed decisions or to otherwise participate in the community. By providing software as a tool to translate and dub the videos into the community's native languages, promoting equitable access and inclusion for education to people regardless of their language. This system is meaningful especially in a highly divergence in regards to language, in order to provide beneficial information and engagement to help individuals, to enhance and contribute towards positive well-being and empowerment. The software will also employ technology, to streamline an accurate translation, making this software a favorable tool for the intended audience who are generally educators, health officials, or community leaders, as a way to help promote awareness on a critical issues. India is a linguistic diversity country, with more than 270 languages. English is the primary language of higher education, which results in a

shortage of educational resources in regional languages. This situation marginalizes a large segment of society. Recently, efforts have been made to provide educational materials in regional languages. The software supports voice translation from video to audio. The main motto of the system is to reach to diverse areas of the country and educate students about the importance of raspberry pi and raspberry pi built systems. The website used here is CIPAM to educate and train students in regional languages about the importance of raspberry pi. And there are still problems with human interaction. Those Google Assistant problems, in order to Communicate with users, they must have access to Wi-Fi and the internet also they are using very limited language so users can get many problems in using them. Google Assistant is a feature of Android smartphones. However, Internet connections are required for this application to function [6].

### Need for Video Dubbing

Video dubbing is the process of replacing the original dialogue in a video with a translated version in a different language. This involves recording new audio tracks with voice actors who speak the target language and syncing this new dialogue with the original video to ensure it matches the lip movements and timing of the characters. Dubbing educational resources into various languages improves student learning by allowing students to access the material in their first language, enabling comprehension and retention. For creators and businesses, dubbing allows their content to reach wider audiences and increase their market and influence globally. Dubbing also allows non-native speakers, particularly immigrants and refugees, to access important and meaningful information and entertainment, which helps to build a more inclusive society. Dubbing videos lowers language barriers, facilitates cultural exchange, and provides equal access to helpful information and entertainment. Dubbing enables viewers who speak different languages to engage with the content while still retaining a cultural context that makes the content relatable and engaging for the local audience. The primary motivation for dubbing video content in Africa (Uganda), South Korea, and India is to expand access and remain relatable while addressing any prospective language barriers. Dubbing in African contexts and Uganda allows for the achievement of many linguistic groups, so that educational, informational, and entertainment content can be made accessible to persons who speak regional languages. Dubbing in South Korea and India also helps increase access while simultaneously retaining all of the cultural components that make the content relevant and engaging for the local audience. Dubbing will take educational materials and translate them into the students' first language; this improves comprehension and guarantees equitable access to resources for students who may not speak the dominant language. Inclusivity of students' native language will also promote equity in education and improve student engagement. Students are more likely to comprehend and remember new information when it is presented to them in their own language. Furthermore, dubbing accurately conveys cultural nuances, facilitating improved relations to understand the material. Finally, dubbing helps with support for language learning and expansion, as pronunciation and examples of usage are provided in the students' first language. Overall, video dubbing narrows the linguistic divide between students and, thus, supports a more effective and inclusive education.

## II. RELATED WORK

More recent developments in deep learning have dramatically transformed the state of the art in multilingual speech processing, with less natural, more scalable, and more cross-lingual voice conversion and dubbing of multimedia content. The pioneering works on deep-learning-based voice conversion provided the basis of speaker and language adaptation, making use of neural networks to simultaneously convert source speech attributes into target speaker attributes to increase cross-lingual naturalness and speaker-similarity [1], [2]. These groundbreaking works proved that bottleneck characteristics and deep learning neural networks are efficient cross-lingual voice conversion by separating linguistic information and identity of a speaker, which are one of the requirements of successful applications in multilingual dubbing. Language-specific neural networks with modularized architecture and language-specific output layers are more effectively capturing the phonetic and prosodic difference between languages and producing a better cross-linguistic voice conversion quality [3]. Simultaneously, neural text-to-speech (TTS)-based voice conversion systems that are further enhanced with phonetic posteriorgrams have been investigated to achieve more natural and controllable cross-lingual voice conversion, thus maintaining linguistic and speaker features across languages more accurately [4]. Outside the voice conversion world, the current research has been skewed to end-to-end multilingual dubbing and audio-visual translation of video materials. New large-scale audiovisual systems of dubbing have been proposed with the goal of jointly modelling speech, visual stimuli, and multilingual translation in order to increase the accuracy of lip-syncing and semantic convergence in the dubbed videos [5]. Continuing this trend, VideoDubber has added speech-conscious length control to machine translation, thus better matching translated speech with the time scale of the original video and eliminating a critical barrier to implementation of more realistic dubbing systems [6]. All these advancements reflect on the growing attention paid to the coordination of speech, translation, and timing issues to provide high-quality video dubbing. More recently, enormous language models and duration-sensitive translation methods have been put to use to build end-to-end multilingual automatic dubbing pipelines. When duration based translation is combined with neural speech synthesis the results have been encouraging in terms of maintaining the timing alignment in addition to increasing the naturalness of the speech being dubbed in a variety of languages[7]. Simultaneously, the existing literature on multilingual automatic speech recognition (ASR) and its outstanding surveys have stressed the relevance of shared multilingual representation and transfer-learning paradigms to enhance resilience on both low-resource and high-resource language-based video dubbing systems- a factual point that directly translates to the scalability of educational and awareness-raising videos [8]. Challenges voice conversion and cross linguistic speech synthesis benchmarks have also added standardized evaluation protocols and datasets thus leading to comparative analysis of multilingual systems [9]. Furthermore, recent studies on voice recorder integration and neural speech synthesis have made a big leap to the quality of the waveforms and real-time implementation capability, thereby making it possible to practically apply voice-conversion architectures to modern multimedia platforms [10], [11]. In addition, multimodal and edge-based speech-translation systems possessing state-of-the-art technology have shown how it was possible to provide

multilingual speech-translation and speech-synthesis in real-time and using resource-restrained environments, a feature that is of special interest when it comes to the distribution of educational and awareness videos [12], [13]. In spite of these developments, most of the literature that is available focuses on individual parts of the system, such as voice conversion, speech translation, or text-to-speech, and does not offer a holistic, AI-based multilingual dubbing pipeline that fulfills the requirements of educational and awareness-oriented video material. Conversely, the suggested system aims to incorporate speech recognition, machine-translation, duration-aware alignment, and neural speech-synthesis into one end-to-end set up, which is specifically optimised in multilingual conscious and educational video. Using the recent advancements in cross-lingual representation learning [14] and the modern neural dubbing systems, this framework will aim to deliver scalable, high-quality, and perceptually natural multilingual dubbing, without loss in semantic faithfulness and definite time alignment.

### III. PROPOSED METHODOLOGY

The pilot system to dub awareness videos into various regional languages has an organized workflow designed to facilitate the dubbing process. The system was evaluated to determine the effectiveness of the proposed AI-driven multilingual dubbing framework. A total of 53 awareness and educational videos were selected for testing purposes. These selected videos were typical examples of real-life information used for public awareness and education. They were originally recorded in English and converted to audio files (dubbing) in various regional languages, including Hindi, Kannada, Telugu, Malayalam, and Tamil. An average running time of 2.5 minutes per video was chosen; however, there were some variances in the lengths of these videos to simulate realistic usage scenarios. The 2.5-minute length provided the opportunity for the system to perform all speech processing required to create dubbed output from the original video and was short enough that the processing could occur without difficulty on a device, such as a Raspberry Pi with limited processing power. The selected dataset allowed the opportunity to test the entire workflow for audio extraction, conversion of speech to text, translation, text to speech generation, and audio-video synchronization, providing ample resources to test the robustness and reliability of the AI-driven multilingual dubbing system for real-world awareness video use. In addition, testing with a diverse range of videos with varying lengths and content provided further validation to the use of the AI-driven multilingual dubbing system for disseminating multilingual information across different linguistic environments.

The workflow process starts with designating an awareness video, and then uses Python libraries such as MoviePy to extract the audio track from the video. The audio from the extracted audio track, is then the basis of the dubbing. Next, the spoken content is converted to text using a speech to text engine (example, Google Speech-to-Text API). Once the rough transcriptions take place, the text gets translated into multiple regional languages (Hindi, Kannada, Telugu, Malayalam, and Tamil) using machine translation engines (example, Google Translate API, Microsoft Azure Translator, and for Tamil our in-house translation from English). Once the translation process is finished, text option ties are generated using Google Text-to-Speech (GTTS). These tools produce

natural sounding voiceovers to the target languages and produce agreeable dubbing.

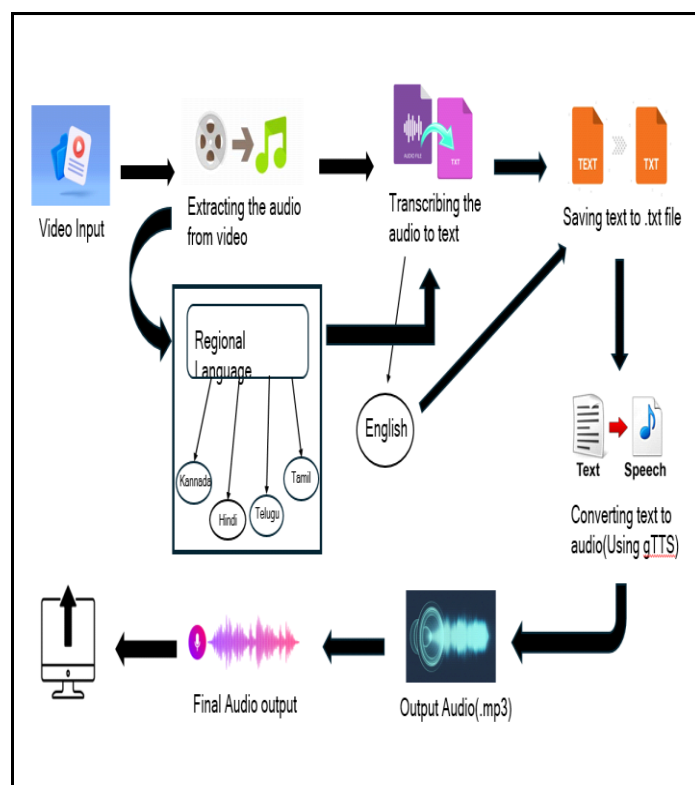


Fig 1. The Proposed system

To make sure the new audio is in sync with the visuals in the video, audio alignment models are used to time the speech with the speaking lip movements and gestures in the video. To perform the audio synchronization the libraries, PYDUB and FFMPEG are used to work with concatenate, process and export the final video output. In an effort to improve portability and costs, the entire software system has been implemented on a Raspberry Pi. This device can read the video, perform the required processing, and play the final dubbed output through an SD card and connected speakers. This interface would allow users to upload videos, select their preferred target languages, and play the dubbed versions conveniently. The system offers an integrated and efficient solution for automatically dubbing awareness videos into multiple regional languages. It ensures accurate translation and natural voice generation and also promotes accessibility by delivering essential content to audiences in their native languages. The entire system focuses on the user's ease of use, understanding, gaining knowledge and gaining awareness in their regional languages. Which helps educating people worldwide with good knowledge. And this ends the proposed methodology with a proposed architecture as shown in Fig. 1. The system was evaluated to determine the effectiveness of the proposed AI-driven multilingual dubbing framework. A total of 53 awareness and educational videos were selected for testing purposes. These selected videos were typical examples of real-life information used for public awareness and education. They were originally recorded in English and converted to audio files (dubbing) in various regional languages, including Hindi, Kannada, Telugu, Malayalam, and Tamil. An average running time of 2.5 minutes per video was chosen; however, there were some variances in the

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#### IV. RESULTS AND DISCUSSION

The system successfully automated the process of dubbing and subtitle creation for awareness videos with Python. The system extracts audio from the input video, converts it into a compatible format, and utilizes Google Speech Recognition for transcription of the spoken text to written text.

TABLE I. SYSTEM SUMMARY

| Description                                              | Tools/Technologies Used                             | Output                                 |
|----------------------------------------------------------|-----------------------------------------------------|----------------------------------------|
| Video Input                                              | Input video file                                    | Original awareness video               |
| Extract Audio from Video                                 | Video processing tools (e.g., MoviePy)              | Audio file                             |
| Transcribe Audio to Text                                 | Speech-to-text engine (e.g., Google Speech API)     | Transcript (.txt)                      |
| Save Transcript                                          | File system                                         | Text file (e.g., transcript.txt)       |
| Translate Text into Regional Languages                   | Translation API/ML models                           | Translated text (Hindi, Kannada, etc.) |
| Convert Translated Text to Audio (TTS)                   | Text-to-speech (e.g., gTTS – Google Text-to-Speech) | Audio files in target language         |
| Synchronize Audio with Original Video (Lip-Sync, Timing) | Audio-video editing tools                           | Dubbed video with synced audio         |

The text is then translated into Kannada using Google Translate, and subtitles are created for the video frames. Furthermore, the translated text is converted into seamless natural-sounding Kannada speech using Google Text-to-Speech (GTTS), producing audio outputs in both MP3 and WAV formats. The output shows a Kannada-dubbed video with subtitles that were in sync with the audio, demonstrating that the system can combine different methodologies (speech recognition, translation, and text-to-speech) into a single workflow. This process significantly minimizes the amount of manual effort required to create a multilingual video while helping to expand the accessibility of information to the larger audience by giving translations in sight and sound. In

summary, the system demonstrates the potential final product of using several AI-related components for real-time multilingual dubbing applications.

TABLE II. PERFORMANCE METRICES FOR PROPOSED SYSTEM

|                                 | Modality             | Methodology                                                       | Limitations    |
|---------------------------------|----------------------|-------------------------------------------------------------------|----------------|
| Yan Wu et al., 2022 [4]         | Text + Speech        | Length-Aware Machine Translation                                  | Data-Dependent |
| Yutian Zhou et al., 2020 [5]    | Video + Audio        | Audio-Visual Deep Learning                                        | Resource-Heavy |
| Hyeonseung Won et al., 2025 [6] | Speech-to-Speech     | Large Language Model                                              | High Cost      |
| Proposed Model                  | Video + Audio + Text | Automatic Speech Recognition, Machine Translation, Text-to-Speech |                |

The above Table II represents a comparative analysis of the existing approaches with our proposed method based on modality, methodology and limitations

TABLE III. PERFORMANCE METRICES FOR PROPOSED SYSTEM

| Language  | WER  | BLEU | MOS |
|-----------|------|------|-----|
| Hindi     | 15.7 | 42.7 | 4.3 |
| Kannada   | 18.5 | 38.7 | 3.9 |
| Telugu    | 17.7 | 36.9 | 4.1 |
| Tamil     | 15.5 | 40.3 | 4.4 |
| Malayalam | 19.9 | 34.7 | 3.9 |

Where Table III summarizes the combination of all the Speech-to-Text (WER), Bilingual Evaluation Understudy (BLEU), and text to Speech (MOS) across the supported regional languages. The combined measurements would allow to provide a direct comparison of the accuracy, translation fidelity, and naturalness which intuitively would show that Hindi and Tamil tend to show better performance in all three cases but Malayalam does show a relatively high WER and a relatively lower MOS which could be explained by the fact that the phonetics of this language is a complicated affair.

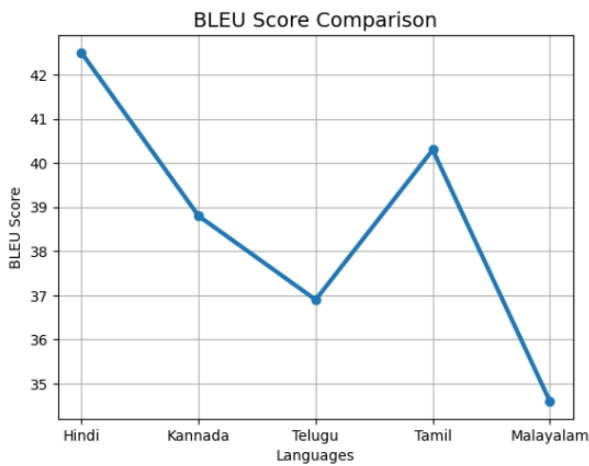


Fig 2. The Performance Metric Bilingual Evaluation Understudy

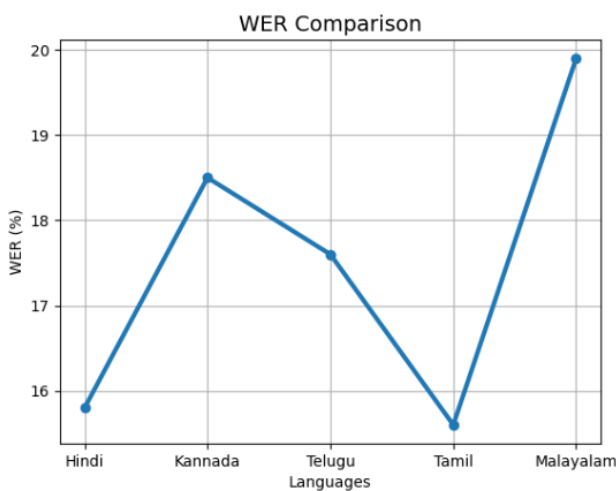


Fig 3. The Performance Metric Word Error Rate

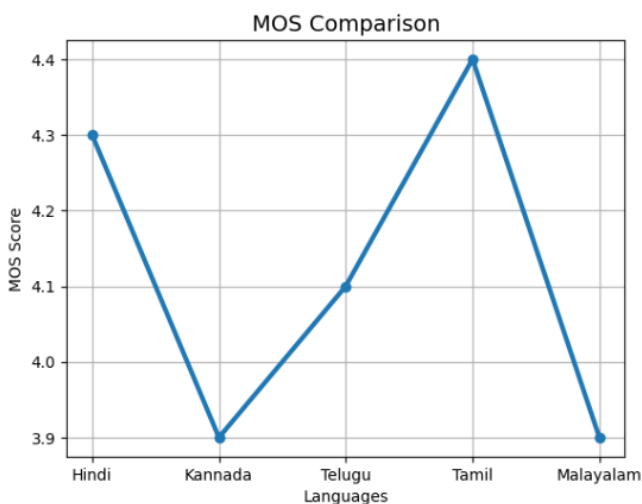


Fig 4. The Performance Metric Mean Opinion Score

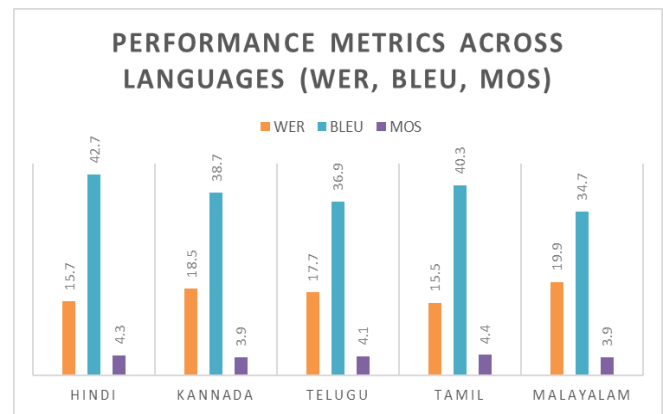


Fig 5. Overall Performance Metrics (WER, BLEU, MOS)

Fig. 2 provides a comparative perspective of BLEU scores by the different Indian languages that have been analyzed highlighting discrepancies in translation efficiency, and demonstrating that select languages are always more likely to reach greater concordance with reference output. The error rate comparison of Word Error rate (WER) in Fig. 3 shows varying recognition error and error rate across the languages as well as shows evident language specific trends in the strengths of the system. The comparisons of Mean Opinion Score (MOS), a measure of user satisfaction and subjective perceptual quality, are provided in Fig. 4, which shows a substantial difference in the degree of perceptual preferences among languages. In Fig. 5, the WER, BLEU, and MOS measurements have been clustered into a single visualization therefore, allowing a holistical cross-metric analysis allowing an integrated comprehension of the objective and the subjective measurement indicators. It summarizes the evidence of the evaluation in a tabular form providing a structured description of the multilingual performance tendencies and supporting the inter-linguistic variability identified. Summing them up, the results indicate consistent cross-linguistic performance gaps and the need to balance performance of multilingual systems by means of language-specific adjustment and optimization. Fig. 5 is provided to give a composite bar graph of WER, BLEU and MOS of all the languages. The analysis makes it obvious to visualize cross-language differences wherein WER fluctuating towards minus in Hindi, BLEU fluctuating towards maxima radiating Hindi and Tamil, and the MOS rating steadily being high in Tamil. The combination of this chart provides an easy to understand, big picture approach to the end-to-end multilingual dubbing pipeline.

## V. CHALLENGES

One of the major challenges encountered during prototype development was achieving accurate speech-to-text and text-to-speech conversion across multiple languages. The differences in pronunciation, accent, and sentences made it difficult to deliver accurate translations and subtitles. Another major challenge encountered was achieving close synchronization between the dubbed speech and the lip movements of the speakers. This resulted from using the limited processing power available from a Raspberry Pi. Additionally, another issue faced was the challenge of getting



the quality of sound and the "real-ness" of the presentation of the voices to be satisfactory in the dubbing. Lastly, the noise or background from and on the original recording of the software prevented using AI to cleaning it up to be a clean sounding natural voice canonized Lexicon speech. Changes to pitch, tempo, and emotional tone can make voices sound less synchronic and nonhuman. Balancing tone and emotion is particularly important for videos meant to promote awareness because the voice should sound sympathetic and engaging rather than flat and robotic. Additionally, it is important to maintain privacy of information and to be responsible in the use of AI. The intention was to generate a prototype that was not overly complicated, but which was as simple and effective as possible. Same again for simplicity's sake. Reducing the wait time for translation and dubbing without breaking the bank is also a key consideration too esp. in rural or resource-poor areas that have limited hardware and internet capabilities. The real test of whether this AI document dubbing system is effective for large-scale awareness systems will be the scalability of the solution, which involves supporting multiple languages and multiple translations.

## VI. CONCLUSION

The approach in dubbing awareness videos into regional languages showcases a viable and scalable solution for dismantling communication barriers within multilingual communities. It demonstrates how the capabilities of speech recognition, machine translation, and text-to-speech synthesis can be combined into a small, cost-effective Raspberry Pi computer to illustrate how AI-type applications can even function on inexpensive hardware. The model captures the complete process of dubbing the awareness videos that automatically translates them to different languages, eliminating the need for more costly and labor-intensive human translation services. Overall, the complete solution is cost-effective which is advantageous in rural or under-resourced communities. It has particular value for government agencies, educational institutions, and non-profit organizations that are interested in cost-efficient ways of disseminating localized awareness content to remote communities. Trial-based testing has shown that the prototype can function reliably with multiple Indian languages while keeping acceptable clarity and accuracy levels. Because the raspberry-pi device is portable, it can be easily used for deployment in low-resource contexts. This expands the systems availability across Asia and Africa where differences in language is often a barrier to sharing information. With additional work, the AI powered multilingual dubbing framework can play a meaningful role in bridging the communication gap and enhance inclusion by facilitating seamless content localization for diverse linguist audiences around the world.

## REFERENCES

- [1] Walczyna and Z. Piotrowski, "Overview of Voice Conversion Methods Based on Deep Learning," *Applied Sciences*, vol. 13, no. 5, p. 3100, 2023. [Online]. Available: <https://doi.org/10.3390/app13053100>
- [2] M. K. Reddy and K. S. Rao, "DNN-Based Cross-Lingual Voice Conversion Using Bottleneck Features," *arXiv preprint, arXiv:1909.03974*, 2019. [Online]. Available: <https://arxiv.org/abs/1909.03974>
- [3] Y. Zhou, X. Tian, E. Yilmaz, R. K. Das, and H. Li, "A Modularized Neural Network with Language-Specific Output Layers for Cross-Lingual Voice Conversion," *arXiv preprint, arXiv:1910.00496*, 2019. [Online]. Available: <https://arxiv.org/abs/1910.00496>
- [3] S. Zhao, H. Wang, T. H. Nguyen, and B. Ma, "Towards Natural and Controllable Cross-Lingual Voice Conversion Based on Neural TTS Model and Phonetic Posteriorgram," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021. [Online]. Available: <https://resourcecenter.ieee.org/conferences/icassp-2021/spsicassp21vid0294>
- [4] Yan Wu, Jing Guo, Xu Tan, Chang Zhang, Bei Li, Rui Song, Lei He, Shujie Liu, Jinyu Li, Sheng Zhao, and Arul Menezes, "VideoDubber: Machine Translation with Speech-Aware Length Control for Video Dubbing," *arXiv preprint, arXiv:2211.16934*, 2022. [Online]. Available: <https://speechresearch.github.io/videodubbing/>
- [5] Yutian Zhou, Ben Shillingford, Yannis Assael, Mohammad Norouzi, Felix de Chaumont, and Demis Hassabis, "Large-Scale Multilingual Audio-Visual Dubbing," *arXiv preprint, arXiv:2011.03530*, 2020. [Online]. Available: <https://arxiv.org/abs/2011.03530>
- [6] Hyeonseung Won, Dain Jeong, Hyunwoo Choi, and Joon Son Kim, "End-to-End Multilingual Automatic Dubbing via Duration-Based Translation with Large Language Models," in *Proc. Conference on Empirical Methods in Natural Language Processing (EMNLP) – Demonstrations*, 2025. [Online]. Available: <https://aclanthology.org/2025.emnlp-demos/37/>
- [7] Hemant Yadav and Shachi Sitaram, "A Survey of Multilingual Models for Automatic Speech Recognition," in *Proc. Language Resources and Evaluation Conference (LREC)*, pp. 5057–5067, 2022. [Online]. Available: <https://aclanthology.org/2022.lrec-1.542/>
- [8] Yi Zhao, Wei-Chen Huang, Xulong Tian, Junichi Yamagishi, Tomoki Toda, Ricard Marxer, and Andreas Strom, "Voice Conversion Challenge 2020: Intra-Lingual Semi-Parallel and Cross-Lingual Voice Conversion," *arXiv preprint, arXiv:2008.12527*, 2020. [Online]. Available: <https://arxiv.org/abs/2008.12527>
- [9] Tamás Viktor Nosek, Stefan Bojan Suzić, Dragan Jovan Pekar, Radiroje Jovan Obradović, Marko Sava Sečujski, and Vladimir D. Delić, "Cross-Lingual Neural Network Speech Synthesis Based on Multiple Embeddings," *International Journal of Interactive Multimedia and Artificial Intelligence (IJIMAI)*, vol. 7, no. 2, pp. 38–47, 2021. [Online]. Available: <https://ijimai.researchcommons.org/ijimai/vol7/iss2/5/>
- [10] Awais Munir, Hamza Nasir, Muhammad Sher, and Muhammad Ahmad Arbab, "Voice Cloning and Synthesis Using Deep Learning: A Comprehensive Study," *International Journal of Innovative Science and Technology (IJIST)*, 2023. [Online]. Available: <https://journal.50sea.com/index.php/IJIST/article/view/1551>
- [11] Yan Zhao, Zhen Huang, and Weicong Li, "Continuous Vocoder Applied in Deep Neural Network Based Voice Conversion," *Multimedia Tools and Applications*, vol. 78, pp. 14673–14691, 2019. [Online]. Available: <https://link.springer.com/article/10.1007/s11042-019-08198-5>
- [12] Muhammad Ameer Khan and Syed Awais Hasan, "Real-Time Speech Translation for Wearable Devices: A Multi-Modal Approach Using Edge Computing and Neural Machine Translation," *International Journal for Research in Applied Science and Engineering Technology (IJRASET)*, vol. 13, 2025. [Online]. Available: <https://doi.org/10.22214/ijraset.2025.74417>
- [13] Mirza Moiz Baig, Ketki Nitesh Butale, Harsh Bandu Meshram, Komal Ravindrarao Barwat, and Anuj Bhasarkar, "An Automated Video Language Translator Using STT-TTT-TTS Translation," *International Journal for Research in Applied Science and Engineering Technology (IJRASET)*, vol. 13, 2025. [Online]. Available: <https://doi.org/10.22214/ijraset.2025.69786>
- [14] Alexis Conneau, Alexei Baevski, Ronan Collobert, Abdelrahman Mohamed, and Michael Auli, "Unsupervised Cross-Lingual Representation Learning for Speech Recognition," *arXiv preprint, arXiv:2006.13979*, 2020. [Online]. Available: <https://arxiv.org/abs/2006.13979>